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|---|---|---|---|---|---|
| 1 | (a) | (i) | **1.** $F = Gm_1m_2/x^2$ | | |
| | | | $= (6.67 \times 10^{-11} \times 2.50 \times 5.98 \times 10^{24})/(6.37 \times 10^6)^2$ | M1 | |
| | | | = 24.6 N (*accept 2 s.f. or more*) | A1 | [2] |
| | | | **2.** $F = mx\omega^2$ **or** $F = mv^2/x$ **and** $v = \omega x$ *(accept x or r for distance)* | C1 | |
| | | | $= 2.50 \times 6.37 \times 10^6 \times (2\pi/24 \times 3600)^2$ | | |
| | | | = 0.0842 N (*accept 2 s.f. or more*) | A1 | [2] |
| | | (ii) | reading = 24.575 – 0.0842 | B1 | |
| | | | = 24.5 N (*accept only 3 s.f.*) | A1 | [2] |
| | (b) | | gravitational force provides the centripetal force | M1 | |
| | | | gravitational force is ‘equal’ to the centripetal force | | |
| | | | (*accept* $Gm_1m_2/x^2 = mx\omega^2$ **or** $F_C = F_G$) | M1 | |
| | | | ‘weight’/sensation of weight/contact force/reaction force is difference between $F_G$ and $F_C$ which is zero | A1 | [3] |